

1

$$\lim_{x \rightarrow \infty} \frac{\ln(e^{2x+5})}{x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{2e^{2x+5}}{e^{2x+5}} = \underline{\underline{2}}$$

2

$$f'(x) = f(x) \cdot (\ln|f(x)|)'$$

$$\ln(\cot x) + x \cdot \frac{1}{\cot x} \cdot \left(-\frac{1}{1+x^2}\right)$$

$$\frac{f'(\sqrt{3})}{f(\sqrt{3})} = \ln \frac{\pi}{2} - \frac{3\sqrt{3}}{2\pi}$$

//

3

$$\frac{dy}{dx} = - \frac{2x+3y}{3x+2y} \Big|_{(x=1, y=1)} = -1$$

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = - \frac{(2+3y')(3x+2y) - (2x+3y)(3+2y')}{(3x+2y)^2}$$

$$\frac{2}{5}$$

//

4

$$\lim_{x \rightarrow 0^+} \frac{x}{\sqrt{3x^2+3x+1}} = 0$$

$$\therefore \lim_{x \rightarrow 0^-} a \cos x = \underline{\underline{a=0}}$$

//

5

$$f'(x) = \sqrt{2} \cos x - 2 \sin^2 x + 2 \cos^2 x + \sqrt{2} \sin x$$

$$= \sqrt{2} (\cos x + \sin x) \{ \sqrt{2} (\cos x - \sin x) + 1 \}$$

$$\therefore \cos x + \sin x = 0 \quad \text{or} \quad \sin x - \cos x = \frac{1}{\sqrt{2}}$$

$$i) \cos x + \sin x = 0$$

$$x = n\pi - \frac{\pi}{4} \quad (n \in \mathbb{Z}) \quad \therefore f\left(n\pi - \frac{\pi}{4}\right) = -3$$

$$ii) \sin x - \cos x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow (\sin x - \cos x)^2 = 1 - 2 \sin x \cos x = \frac{1}{2}$$

$$\therefore \sin x \cos x = \frac{1}{4}$$

$$\Rightarrow f = \sqrt{2} (\cos x - \sin x) + 2 \cos x \sin x = \frac{3}{2}$$

$$\therefore M = \frac{3}{2}, \quad m = -3 \quad \Rightarrow \underline{\underline{M+m = -\frac{3}{2}}}$$

//

6

$$\frac{a}{a+1} \rightarrow 0 \quad \dots \quad \frac{b}{b+1} \rightarrow 0 \quad \therefore \underline{\underline{b=1}}$$

//

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{ax+1} - 1}{x} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{1}{3} (ax+1)^{-\frac{2}{3}} \cdot a = \frac{a}{3} = \frac{1}{3}$$

$$\therefore a = \frac{3}{2}$$

$$\Rightarrow \underline{\underline{a+b = \frac{5}{2}}}$$

//

7

$$(f^{-1})'(1) = \frac{1}{f'(0)} = \underline{\underline{\frac{1}{2}}}$$

//

$$f'(x) = 14e^x \Big|_{x=0} = 2 \quad \uparrow$$

8

$$f: x^3 - y^3 - 2 = 0, \quad x=1 \Rightarrow y=-1$$

$$\frac{dy}{dx} (=y') = - \frac{3x^2}{-3y^2} = \frac{x^2}{y^2} \Big|_{(1,-1)} = 1$$

$$\therefore y'' = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{2xy^2 - x^2 \cdot 2y \cdot y'}{(y^2)^2} \Big|_{(1,-1)} = 4$$

9

sol I)

$$\rho \lim_{x \rightarrow \infty} x \cdot \ln \left(\frac{x+2a}{x-1} \right) = \rho \lim_{x \rightarrow \infty} \frac{\ln(x+2a) - \ln(x-1)}{\frac{1}{x}}$$

$$\stackrel{0}{=} \rho \lim_{x \rightarrow \infty} \frac{\frac{1}{x+2a} - \frac{1}{x-1}}{-\frac{1}{x^2}} = \frac{-1-2a}{(x+2a)(x-1)}$$

$$= e^{1+2a} = e^{2.3} \quad \therefore a=1.011$$

sol II)

$$\lim_{x \rightarrow \infty} \left(\frac{x+1+2a}{x-1} \right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{1+2a}{x-1} \right)^{\frac{x-1}{1+2a} \cdot \frac{(1+2a)}{x-1} \cdot x}$$

$$= e^{1+2a} = e^{2.3} \quad \therefore a=1.011$$

10

 $x=0$ 에서 연속성 조사!

$$\lim_{x \rightarrow 0^+} \left((1+x)^{\frac{1}{x}} + ax + b \right) = e + b \quad \therefore b = -e \text{ \& } c = 0$$

$$\lim_{x \rightarrow 0} |x|^{\frac{3}{2}} \sin \frac{1}{x} = 0 \quad \left(\because \left| |x|^{\frac{3}{2}} \sin \frac{1}{x} \right| \leq |x|^{\frac{3}{2}} \right)$$

$\downarrow x \rightarrow 0^- \quad \downarrow x \rightarrow 0^-$
($\therefore \Delta x$ 2 경위)

과도분계수 = 무한계수 조사!

$$\lim_{h \rightarrow 0} \frac{|h|^{\frac{3}{2}} \sin \frac{1}{h}}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^{\frac{1}{h}} + ah - e}{h}$$

$\parallel \frac{0}{0} \quad \frac{1}{h} - \frac{1}{4h}$

$$\lim_{h \rightarrow 0} \frac{a + (1+h)^{\frac{1}{h}} \left(-\frac{1}{h^2} \ln(1+h) - \frac{1}{4h^2} \right)}{h}$$

$$\ln(1+h) \stackrel{\text{Mac}}{=} h - \frac{h^2}{2} + \dots$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{a + e \left(-\frac{1}{h^2} \left(h - \frac{h^2}{2} \right) + \frac{1}{h} - 1 \right)}{h} = 0$$

$$\therefore a - \frac{e}{2} = 0 \Rightarrow a = \frac{e}{2}$$

$$\therefore a+b+c = -\frac{e}{2}$$

11

$$\lim_{x \rightarrow 0} \frac{\sin(\sqrt{x^2+x} - 2)}{\cos(xe^x) - 1} \cong \sqrt{x^2+x} - 2 = \frac{x^2}{\sqrt{x^2+x} + 2}$$

$$= \frac{1}{4} \lim_{x \rightarrow 0} \frac{x^2}{\cos(xe^x) - 1}$$

sol I)

$$\stackrel{0}{=} \frac{1}{4} \lim_{x \rightarrow 0} \frac{2x \cdot e^x}{-\sin(xe^x) (e^x + xe^x) \cdot e^x} = -\frac{1}{2}$$

sol II)

$$e^x \stackrel{\text{Mac}}{=} 1 \quad \Rightarrow \frac{1}{4} \lim_{x \rightarrow 0} \frac{x^2}{\cos x - 1} = -\frac{1}{2}$$

$\parallel \text{Mac}$
 $1 - \frac{1}{2}x^2$

12

$$f^{-1}(b) = \frac{1}{f'(a)} = \frac{1}{\frac{1}{2}}$$

13

sol I) 코피탈. ($\therefore$ 음분).

$$\lim_{x \rightarrow 2023} \frac{\frac{f'(x)}{2f'(x)}}{1} = \frac{1}{2} \cdot \frac{3}{3} = \frac{1}{3}$$

sol II) 분자 극한.

$$\lim_{x \rightarrow 2023} \frac{(1-f(x)-3)(f(x)+3)}{(x-2023)(f(x)+3)} = \frac{1}{6} \lim_{x \rightarrow 2023} \frac{f(x)-f(2023)}{x-2023} = \frac{1}{6} f'(2023) = \frac{1}{3}$$

$f(2023)+3=6$

14

$$y' = d^{\sin x} \left(\sin x \cdot \ln|x| \right)' = d^{\sin x} \left(\cos x \cdot \ln|x| + \sin x \cdot \frac{1}{x} \right)$$

$$\text{at } x = \frac{\pi}{2}, \quad \frac{\pi}{2} \cdot \left(\frac{1}{\pi} \right) = 1$$

15

$$\lim_{x \rightarrow 0} \frac{g(e^x) - g(1)}{e^x - 1} \cdot \frac{1}{(e^x + 1)} \stackrel{e^x \rightarrow 1}{=} \lim_{t \rightarrow 1} \frac{g(t) - 1}{t - 1} \cdot \frac{1}{t + 1}$$

 $= \frac{1}{2} g'(1)$ 를 구하자!

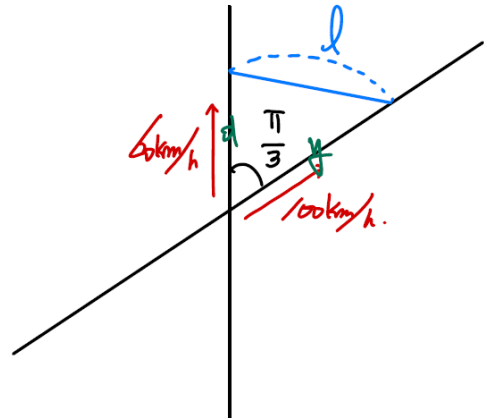
$$g(1) = g(f^{-1}(0)) = 6 \cdot \frac{1}{f'(2)} = \frac{6}{e^2} \Rightarrow g'(1) = \frac{3}{e^2}$$

16

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^0 - 1}{e^0 + 1}$$

$$\therefore \lim_{x \rightarrow 0} \left(\frac{1 - \frac{e^0 - 1}{e^0 + 1}}{1 + \frac{e^0 - 1}{e^0 + 1}} \right) = \lim_{x \rightarrow 0} \frac{2}{2e^0} = 1$$

17



$$\frac{dx}{dt} = 60 \quad \frac{dy}{dt} = 100$$

$$l^2 = x^2 + y^2 - 2xy \cos \frac{\pi}{3} = x^2 + y^2 - xy$$

$$2l \cdot \frac{dl}{dt} = 2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} - y \cdot \frac{dx}{dt} - x \cdot \frac{dy}{dt}$$

$$(30 \text{ 분 } \text{ } \text{ } : x=30, y=50)$$

$$l = \sqrt{1900} = 10\sqrt{19}$$

$$20\sqrt{19} \cdot \frac{dl}{dt} = 60 \cdot 60 + 100 \cdot 100 - 50 \cdot 60 - 30 \cdot 100$$

$$\Rightarrow \frac{dl}{dt} = 20\sqrt{19}$$

18

$$\frac{dy}{dx} = \frac{2t - t^2}{e^{2t}}$$

$$\begin{aligned} \frac{d}{dx} \left(\frac{dy}{dx} \right) &= \frac{\frac{dt}{dx} \cdot \frac{d}{dt} \left(\frac{dy}{dx} \right)}{e^{-2t}} \\ &= (2-2t)e^{-2t} - 2(2t-t^2)e^{-2t} \\ &= (2t^2 - 6t + 2)e^{-2t} \\ &= e^{-2t} (2t^2 - 6t + 2) < 0. \end{aligned}$$

$$\therefore \frac{3}{2} - \frac{\sqrt{5}}{2} < t < \frac{3}{2} + \frac{\sqrt{5}}{2}.$$

$$\therefore \underline{t=1.2}$$

19

높이: h . 넓이: S . 길변: d .

$$\frac{dh}{dt} = 1, \quad \frac{dS}{dt} = 2, \quad \underline{h=10, S=100}.$$

$$S = \frac{1}{2}dh$$

$$100 = \frac{1}{2} \cdot d \cdot 10$$

$$\therefore d = 20.$$

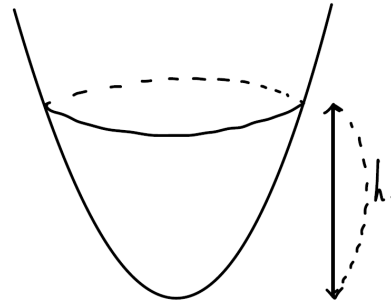
$$\frac{dS}{dt} = \frac{1}{2} \left(h \cdot \frac{dh}{dt} + d \cdot \frac{dh}{dt} \right)$$

$$2 = \frac{1}{2} \left(10 \cdot \frac{dh}{dt} + 20 \cdot 1 \right)$$

$$= 5 \frac{dh}{dt} + 10$$

$$\therefore \underline{\underline{\frac{dh}{dt} = -1.6 \text{ cm/min.}}}$$

20



$$V = \pi \int_0^h z dz$$

$$\leadsto \frac{dV}{dt} = \pi h \cdot \frac{dh}{dt}$$

$$\text{예로 유입되는 물의 양: } q. \Rightarrow \frac{dV}{dt} = \frac{q}{\pi} \cdot \frac{1}{h}$$

$$\therefore \underline{\underline{\frac{dh}{dt} \text{ 는 } \frac{1}{h} \text{ 에 비례}}}$$

21

㉠

 $(x=y=0) \Rightarrow \text{clear!}$
 $(x \neq y) \dots \text{WLOG. } x > y.$
 $\cos \dots$ 연속 & 미가 $\Rightarrow$ MVT에 의해

 $\cos x - \cos y = -\sin c(x-y) \quad \text{인 } c \in (y, x) \text{ 가 존재.}$

$$\therefore |\cos x - \cos y| = |-\sin c(x-y)| \leq |x-y| \quad \square$$

㉡

 $\text{값 불증: } f'' < 0$

$$f''(x) = \frac{1}{4} - \frac{2}{x^3} \dots \frac{1}{x} > 1 \Rightarrow f''(x) < 0 \quad \square.$$

㉢

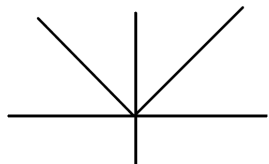
$$f \leq \frac{8}{x} + \sqrt{x} \quad \text{---}$$

$$f' = -\frac{8}{x^2} + \frac{1}{2\sqrt{x}} = -\frac{16}{2x^2} + \frac{\sqrt{x}}{2x^2} = 0$$

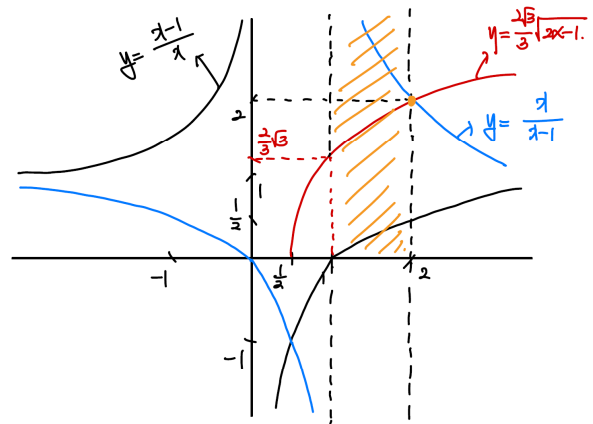
$$\therefore \sqrt{x} = 16 \Rightarrow x = 256$$

$$f(256) < 4 \quad \dots \text{False}$$

㉣


 $\dots \text{clear!}$

22

 $\text{식으 풀 수도 있지만 편의상 그래프 접근!}$

 $\ast$ 교점 구하는 식은 생략...

$$\therefore 1 < x \leq 2 \Rightarrow a+b=3$$

23

$$\lim_{t \rightarrow d} \frac{\sin t - \sin d}{t-d}$$

 $\sin d: \text{연속 & 미분가능} \Rightarrow \text{MVT에 의해 } (g(c) \triangleq \sin c).$

$$\sin t - \sin d = g'(c)(t-d). \quad c \in (d, t).$$

$$\frac{1}{1+c^2} \quad t \rightarrow d: c \rightarrow d$$

$$\therefore \lim_{t \rightarrow d} \frac{\sin t - \sin d}{t-d} = \lim_{c \rightarrow d} \frac{1}{1+c^2} = \frac{1}{1+d^2} = f(d).$$

$$\therefore f(-1) = \frac{-2x}{(1+x^2)^2} \Big|_{x=-1} = \frac{1}{2}$$

 $\ast$ 코피탈 써도 됨($\because$ 응답).

24

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{(\sqrt{x^2+1} - 1)(\sqrt{x^2+1} + 1)}{f(x) \cdot (\sqrt{x^2+1} + 1)} &= \frac{x^2 + 1 - 1}{f(x)} = \frac{x^2}{f(x)} \\
 &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2+1} + 1}
 \end{aligned}$$

$$* \lim_{x \rightarrow 0} x \sin \frac{1}{x} : \left| x \sin \frac{1}{x} \right| \leq |x|$$

$\downarrow x \rightarrow 0$ $\downarrow x \rightarrow 0$
 0 0

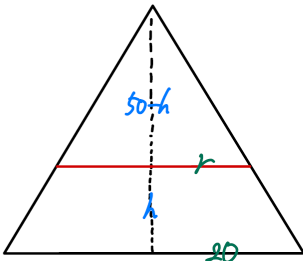
$$\therefore \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0. \quad \therefore f(x): \text{불연속.}$$

$$\therefore \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + f(x)} + 1} = \frac{1}{2}$$

$\downarrow \lim_{x \rightarrow 0} f(x) = 0.$

$$\therefore a_3 = \frac{1}{8}, a_6 = \frac{1}{12} \Rightarrow a_3 + a_6 = \frac{1}{4}$$

25



$$\text{삼각비} \Rightarrow 20:r = 50:50-h \Rightarrow r = \frac{2}{5}(50-h)$$

$$\therefore V = \frac{\pi}{3} \times 20^2 \times 50 - \frac{\pi}{3} r^2 (50-h)$$

$$= \frac{\pi}{3} \cdot 20^2 \cdot 50 - \frac{\pi}{3} \cdot \frac{4}{25} (50-h)^3$$

$$\frac{dV}{dh} = \frac{4}{25} \pi \cdot (50-h)^2 \cdot \left(\frac{dh}{dh} \right) = \frac{4}{25} \pi (50-h)^2$$

26

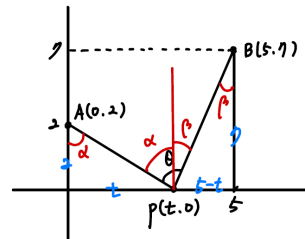
$$|f(x) - 1|^2 = x^2 + 1$$

$$\Rightarrow f(x) - 1 = \pm \sqrt{x^2 + 1} \Rightarrow f(x) = 1 \pm \sqrt{x^2 + 1}$$

$$f(0) = 1 \pm \sqrt{1} = 0 \quad \therefore \text{중간값: } - \Rightarrow f(x) = 1 - \sqrt{x^2 + 1}$$

$$\therefore f'(x) = -\frac{x}{\sqrt{x^2 + 1}} \Big|_{x=1} = -\frac{1}{\sqrt{2}}$$

27



$$\therefore \cos \theta = \cos(\alpha + \beta) = \frac{\cos \alpha \cos \beta - \sin \alpha \sin \beta}{1 - \cos \alpha \cos \beta} = \frac{\frac{t}{5} + \frac{5-t}{2}}{1 - \frac{t}{5} \cdot \frac{5-t}{2}} = \frac{5t+10}{t^2-5t+14} \triangleq f(t)$$

$$\Rightarrow f'(t) = \frac{(-5) \cdot (t^2-5t+14) - (5t+10) \cdot (2t-5)}{(t^2-5t+14)^2} = 0 \quad \therefore t = -2 \pm 2\sqrt{7} \quad (t > 0)$$

$t=0$ & $t=5$ 일때 check! ... 생략!

28

$$\text{또한 } r = \langle 2t^2, 2t \rangle$$

$$\therefore |r| = \sqrt{(2t^2-1)^2 + (2t-4)^2} = \sqrt{4t^4 - 16t + 17} \triangleq f(t)$$

$$f'(t) = 16t^3 - 16 = 0 \quad \therefore t = 1$$

29

$$f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + x^3 + xh^2 - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x^3}{h} + \lim_{h \rightarrow 0} \frac{h(x^2 + xh)}{h}$$

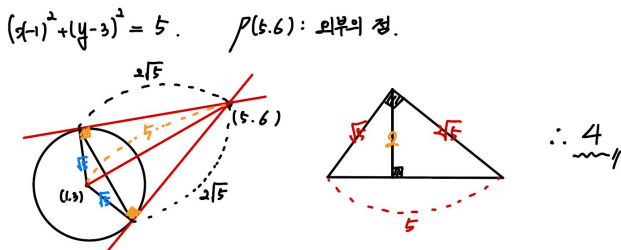
$$= 1 + x^2$$

$$\therefore f(x) = x + \frac{1}{3}x^3 + C \Big|_{f(0)=0} = 0 \quad \therefore C=0 \quad \Rightarrow f(x) = \frac{1}{3}x^3 + x$$

$$\therefore f(3) = 9 + 3 = 12$$

* 편미분을 이용해서도 구할 수 있다 ... 생략!

30

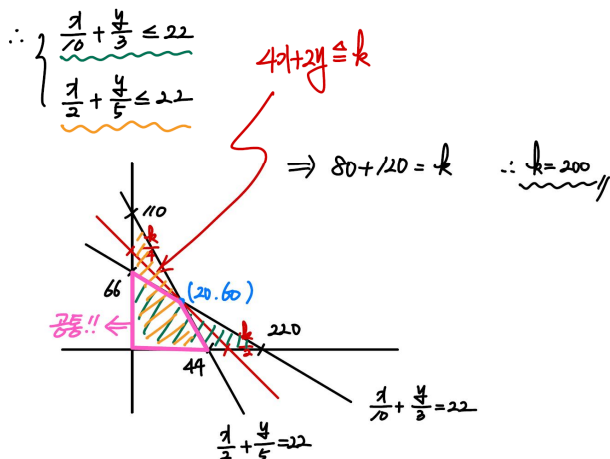


31

S 패변 생산대수: x $\xrightarrow{\text{하사생산}}$

L 패변 " : y $\xrightarrow{\text{하사생산}}$

	기계 M	기계 N
S	$\frac{1}{10}$	$\frac{1}{20}$
L	$\frac{1}{3}$	$\frac{1}{5}$



32

- $f(x) \triangleq \tan^{-1}x \dots$ 연속함수가 MVT에 의해
 $\tan^{-1}x - \tan^{-1}1 = \frac{1}{1+c^2} (x-1), \quad c \in (1,x).$
 $x = \frac{4}{3} : \tan^{-1}x - \frac{\pi}{4} = \frac{1}{1+c^2} \cdot \frac{1}{3}$
 $C : \frac{4}{3} : \tan^{-1}x - \frac{\pi}{4} > \frac{2}{25}, \quad C=1 : \tan^{-1}x - \frac{\pi}{4} < \frac{1}{8}$
 $\Rightarrow \frac{\pi}{4} + \frac{2}{25} < \tan^{-1}x < \frac{\pi}{4} + \frac{1}{8}$
- $f''(x) = -\frac{1}{2}(4x^3)^{-\frac{3}{2}} \cdot 3x^2 = -\frac{3}{2}x^{\frac{1}{2}}(4x^3)^{-\frac{3}{2}}$
 $g''(x) = -\frac{f''(x)}{|f''(x)|^3} = \frac{3}{2}|g''(x)|^2$
- (반례).

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0) \end{cases}, \quad g(x) = x.$$
- clear!